Composing Complex Numerals in Mandarin

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1. Introduction

In this talk, I provide a semantics that is syntactically informed for Mandarin numerals.

The focus of the analysis is not on the meaning of numerals per se, but rather how complex numerals relate to simplex numerals, and how complex numerals are interpreted in numeral-classifier constructions.

❒ First, I show that nuemral items in Mandarin fall into two syntactic classes, Num° and N° (§ 2).

❒ Second, I show how complex numerals can be built in syntax (§ 3).

❒ Last, I provide a semantics of relating complex numerals its building blocks, and their interpretation under numeral-classifier constructions (§ 4).

2. The Syntax of Mandarin Numerals

It is usual for the system of nuemral items in a language to show divided behaviours (Kayne 2019). This is indeed the case in Mandarin.

Table 1 shows the list of numeral items in Mandarin.

❒ I call those that correspond to numbers in $[\![1, 10]\!]$ *small numerals*;

❒ and bases $\{10^k \mid k \in \{2, 3, 4, 8, 12\}\}$ *ten's powers* (excluding ten!).

Small Numerals										Ten's Powers				
yī	*èr*	*sān*	*sì*	*wǔ*	*liù*	*qī*	*bā*	*jiǔ*	*shí*	*bǎi*	*qiān*	*wàn*	*yì*	*zhào*
1	2	3	4	5	6	7	8	9	10	10^2	10^3	10^4	10^8	10^{12}

Table 1: Numeral Items in Mandarin

Small numerals and tens powers display very distinct syntactic properties. The distinction holds across both arithmetical and counting uses.

❒ Ten's powers but not smalll numerals always need the support of of a multiplier to express their base values.

➢ Compare (1) with (2)

❒ Ten's powers but not small numerals, when used as multiplicands in multiplicative complex numerals, can trigger suppletion on 'one' and 'two'.[1]

➢ Compare (3) with (4) (for space reasons I only include examples involving suppletive 'two')

(1) Ten's powers always need the support of a multiplier *yi* 'one' to express their base values.

a. *(yì) {bǎi/ qiān} kē píngguǒ
$\boxed{\text{one}}$ hundred thousand CL apple
'one {hundred/ thousand} apples'

b. *(Yì) {bǎi/ qiān} shì dà shùzì.
$\boxed{\text{one}}$ hundred thousand COP big number
'One {hundred/ thousand} is a big number.'

(2) Small numerals do not need multiplier *yi* 'one' to express their base values.

a. {sān/ shí} kē píngguǒ
three ten CL apple
'{three/ ten} apples'

b. {Sān/ Shí} shì xiao shùzì.
three ten COP small number
'{Three/ Ten} is a small number.'

(3) Ten's powers can trigger suppletion on multiplier 'two'

a. {(?)èr/ {^{OK}**liǎng**} {bǎi/ qiān} kē píngguǒ.
two $\boxed{\text{two.SUPP}}$ hundred thousand CL apple
'two {hundred/ thousand} apples'

b. {(?)Èr/ {^{OK}**Liǎng**} {bǎi/ qiān} shì dà shùzì.
two $\boxed{\text{two.SUPP}}$ hundred thousand COP big number
'Two {hundred/ thousand} is a big number.'

[1] See Wu (2024) for why 'one' also has a suppletive form.

(4) Small numerals do not trigger suppletion on multiplier 'two.

a. {**èr**/ *liǎng} shí kē píngguǒ
two two.SUPP ten CL apple
'twenty apples'

b. {**Èr**/ *Liǎng} shí hé yì bǎi {**yī**/ *yì} shí shì xiǎo shùzì.
two two.SUPP ten COP small number
'Twenty is small numbers.'

Simplex numerals (which can only be small numerals since ten's powers always need the support of a multiplier), or complex numerals (which are built from small nuemrals plus ten's powers) never have predicative use, which speaks against the adjectival analysis.

❒ Regardless of whether there is a copula, numerals never have predicative use in Mandarin (5).

❒ The only way to remedy (5) is to use (existential) 'be', and add a classifier after the numeral (6).

(5) No predicative use of numerals.
*Xuéshēng (shì) {sān/ shí-èr/ liǎng-bǎi}.
student COP three ten-two(=twelve) two-hundred
Intended: There are {three/ twenty/ two hundred} students.

(6) Remedy to the ungrammaticality of (5): to use *yǒu* and add a classifier.[2]
Xuéshēng yǒu {sān/ shí-èr/ liǎng-bǎi} ×(míng).
student be three ten-two(=twelve) two-hundred CL
Intended: There are {three/ twenty/ two hundred} students.

❒ The requirement of a classifier in (6) suggests that the sentence involves movement of NP from [Num Cl NP].

❒ which in turn suggests that all numerals under counting uses come hand-in-hand with classifer-noun's.

❒ We will see in § 3 that all numerals, no matter used arithmetically or in counting, co-occur with classifier-noun's.

Based on these facts I propose that

❒ Small numerals are exclusively functional elements, merging as Num°.

❒ Ten's powers are exclusively N°, which themselves combine with Num° (and Cl°).

[2]The marker × suggests inappropriateness. Without a classifier, (6) may not be too bad as "ungammatical", but makes you sound like you are from 10th century (especially with simplex numerals).

3. Building Complex Numerals in Syntax

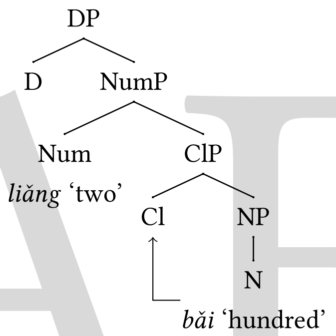
Multiplicative complex numerals fall in two classes.

❒ If the multiplicand numeral is a ten's power, then it merges with the multiplier simply like a classifier-noun merges with Num°.

➢ See (7), where I adopt the Mandarin DP structure by Tang (1990) a.o.. I also assume that ten's powers need to move to Cl° (Simpson, Soh & Nomoto 2011), for reasons I will go through today.

➢ In some sense, 'two hundred' is structurally no different from 'two apples.'

(7) Multiplicative complex numerals, base=small numeral.
liǎng bǎi 'two hundred'

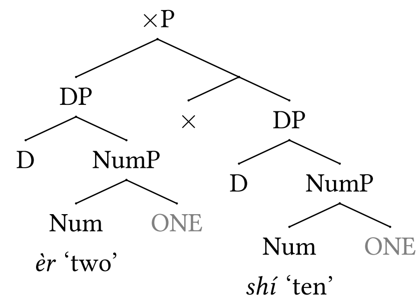


❒ If the multiplicand numeral is a small numeral (which can only be ten), then both the multiplcand and the multplier are DP's.

➢ See (8).

➢ Each numeral takes a silent NP, denoted as ONE, as its complement, much in the spirit of Moltmann (2013).[3]

(8) Multiplicative complex numerals, base=ten's power.
èr shí 'two ten (=twenty)'

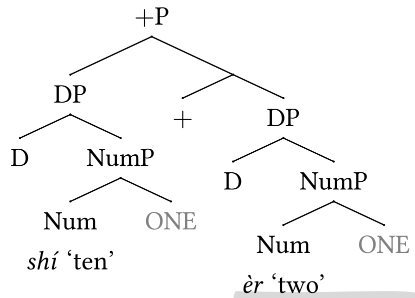


[3]See also Kayne (2005), Collins (2007), Moltmann (2022) i.a. for the notion of light noun.

For additive complex numerals, the added two subparts are always full-fledged DP themselves, with the relating head being $+°$.

- ❒ See (9).
- ❒ It deserves to be thought whether $\times°$ should be collapsed $+°$ into a single head in syntax, with their different semantics's attributed to encyclopædic knowledge.

(9) *shí èr* 'ten two (=twelve)'



Further complex numerals can be built through recursive merger of $\times°$ and $+°$.[4]

Under this analysis, the set of numeral items in Mandarin are split into Num°-kind (small numerals) and N°-kind (ten's powers), each of them has consistent syntactic properties, in the spirit of anti-homophony (2003, 2022).

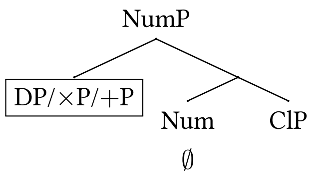
Now I can relate the arithmetic use and the counting use of numerals.

- ❒ For complex numerals, the counting use is simply its counting use merged in Spec,NumP, with the head Num being empty.
- ❒ For simplex numerals, in counting they are the Num° heads. Arithmetically they are DP which involve [Num ONE], which is very desirable since the structure [Num ONE] is now not motivated for only being embedded under complex numerals such as (8, 9), but it has its own independent use.

(10) Complex numerals.

a. Arithmetical use: DP/×P/+P, such as shown in (7, 8, 9)

b. Counting use: DP/×P/+P (in Spec,NumP)



[4] Actually only $+°$, since multiplying an already-complex numeral in Mandarin is not felicitous.

(11) Simplex numerals

a. Arithmetical use: DP i.e. $[_{\mathrm{DP}}[_{\mathrm{NumP}}\ \boxed{\mathrm{Num}}\ \mathrm{ONE}]]$

b. Counting use: Num° taking regular classifier-noun

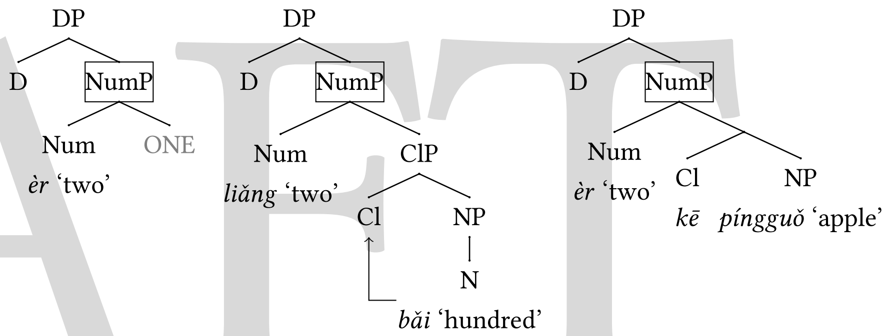
4. The Semantic Side

4.1. Part 1

I argue that no matter it is ten's powers, or silent ONE that small numerals take as complements, they are all semantically similar to nouns. Under this view, sentences like "Two is a prime number" do not involve reference to mathematical numbers, but simply to entities, for Moltmann (2013) but pace Zweig (2005), Ionin & Matuchansky (2018) a.o..

- ❒ The boxed nodes in (12) have parallel semantics.

(12)



I argue that the set of individuals involve abstract entities that are for arithmetical ends.

- ❒ Ten's powers and ONE are predicates for these abstract arithmetical entities, seen (13a) and (13b).
- ❒ There is a tiny difference. The denotation of ONE is the set of already-atomised one's, but for ten's powers, their denotations are yet to be atomised by Cl°.

(13) a. $[\![\mathrm{ONE}]\!] = \textsc{one}(x) \wedge \textsc{atom}(x)$.

b. $[\![\textit{bǎi}]\!] = x.\textsc{hundred}(x)$.

While this talk is not about the semantics of numerals per se, I still have to have one. The semantics of *èr* 'two', for example, has the value in. This semantics is similar to Chierchia (**<empty citation>**).

(14) $[\![\textit{èr}]\!] = \lambda P.\lambda x.\exists X[x = \oplus X \wedge \#X = 2 \wedge \forall x' \in X[P(x') \wedge \textsc{atom}(x)]]$.

The semantics of D° for arithmetically used numerals, as in the first two trees in (12), can simply be a choice function. Here I give an explicit algorithm.

- ❒ Suppose that abstract entities as one's and hundred's are all identifiable by subscripted indices.

- E.g. o_1 and o_{13} are two different one's. h_4 and h_6 are two different hundred's.

- I denote the determiner takes sets of mereological sums of abstract arithmetical entities as $\mathrm{D_n}$.

(15) $[\![\mathrm{D_n}]\!] = \lambda P.\ \mathrm{argmin}_{h \in P} \sum_{\substack{h' \sqsubseteq h \\ \textsc{atom}(h')}} \mathrm{idx}(h'),$

where idx is the function that retrieves the index that a numeric unit carries.

4.2. Part 2

There are two tasks left.

- What is the compositional semantics for complex numerals?

- How should complex numerals be interpreted under numeral-classifier constructions?

I propose that what $\times^\circ$ and $+^\circ$ do is to combine two individuals into a new one, which I represent as vectorisation.

- $\times^\circ$ and $+^\circ$ must do combination differently such that addition and multiplication can be assessed differently.

- I represent this difference by using two ways of vectorisation: $\times^\circ$ vectorises vertically and $+^\circ$ horizontally.

- I let S denote the set of abstract arithmetic entities, this is the domain from which $\times^\circ$ and $+^\circ$ take their arguments.

(16) a. $[\![\times^\circ]\!] = \lambda s_1.\lambda s_2. \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}.$

b. $[\![+^\circ]\!] = \lambda s_1.\lambda s_2. \begin{bmatrix} s_1 & s_2 \end{bmatrix}.$

Definition 1. *Let S_0 be the set of mereological sums of one's, hundred's, thousand's, {ten thousand}'s, {ten million}'s and trillion's. Define*

$$S_i = S_{i-1} \cup \left\{ \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} a & b \end{bmatrix} : a, b \in S_{i-1} \right\}.$$

Then define

$$S = \lim_{i \to +\infty} S_i.$$

I need to make one clarification.

- Using vectors does not mean that they are not entities. It is just a matter of representation.

- I can as well exploit operation symbols like $\boxplus$ or $\boxtimes$, or a collapse function (similar to $\uparrow$ in Landman 1989). But this needs to be handled seriously for logical reasons.

- Therefore, the vector representation is adopted here.

The last thing to do is to figure our how to interpret numeral-classifier constructions involving complex numerals, e.g. the structure in (10b).

- We need a semantics for the head Num° that relates a complex numeral (which has a entity interpretation) and a ClP (which is a set of atoms).

- The trick is to come up with a function that can read the elements in S and returns the corresponding number.

- Essentially, the denotation of Num° can be built from the denotation of a small numeral.

- Compare (17) and (18). What Num° does is to take one more argument (s) than *èr* 'three', and replace 2 with $f(s)$.

(17) $[\![\grave{e}r]\!] = \lambda P.\lambda x.\exists X[x = \oplus X \wedge \#X = 2 \wedge \forall x' \in X[P(x') \wedge \textsc{atom}(x)]].$

(18) $[\![\mathrm{Num}_\emptyset]\!] = \lambda P.\lambda x.\underline{\lambda s}.\exists X[x = \oplus X \wedge \#X = \underline{f(s)} \wedge \forall x' \in X[P(x') \wedge \textsc{atom}(x)]].$

Definition 2. *Define $f : S \to \mathbb{N}^*$, where*

$$f(s) = \begin{cases} f(s_1)f(s_2) & \exists s_1, s_2 \in S \textit{ such that } \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = s \\ f(s_1) + f(s_2) & \exists s_1, s_2 \in S \textit{ such that } \begin{bmatrix} s_1 & s_2 \end{bmatrix} = s \\ n & \exists X \subseteq \textsc{one} \textit{ such that } s = \bigoplus X \wedge \#X = n \wedge \forall x \in X[\textsc{atom}(x)] \\ 100n & \exists X \subseteq \textsc{hundred} \textit{ such that } s = \bigoplus X \wedge \#X = n \wedge \forall x \in X[\textsc{atom}(x)] \\ \vdots & \vdots \end{cases}.$$

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